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Education

Leibniz University Hannover

Ph.D., Mathematics Mar. 2023 – today
Advisor: Stefan Schreieder

Leibniz University Hannover

Master of Science, Mathematics Oct. 2020 – Nov. 2022
Thesis: “Rationality questions and decomposition of the diagonal”
Advisors: Stefan Schreieder and Nebojsa Pavic

Leibniz University Hannover

Bachelor of Science, Mathematics Oct. 2017 – Aug. 2020
Thesis: “Intersection theory and enumerative geometry” (German)
Advisors: Stefan Schreieder

Research Interests

Algebraic cycles, rationality problems, degenerations of algebraic varieties, geometry in positive characteristic.

Publications and Preprints

4. *Torsion order and irrationality of complete intersections* (with Guoyun Zhang),
arXiv preprint [2510.24258](#), submitted.
3. *On the rationality problem for low degree hypersurfaces* (with Stefan Schreieder),
arXiv preprint [2409.12834](#), to appear in Forum of Math. Pi.
2. *On zero-cycles of varieties over Laurent fields*,
arXiv preprint [2402.14406](#), submitted.
1. *The Diagonal of $(3,3)$ fivefolds* (with Bjørn Skauli), Ann. Inst. Fourier, Online first, 31 p.
DOI [10.5802/aif.3735](#), arXiv [2303.00562](#).

Talks

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| “Torsion order and irrationality of complete intersections”
Humboldt University Berlin, AG seminar, Berlin | December 2025 |
| “On the rationality problem for hypersurfaces” (mini-course)
SCMS Algebraic Geometry Summer School, SCMS, Shanghai. | July 2025 |
| “Cubic fourfolds of discriminant 24 and rationality (after B. Hassett)”
Leibniz University Hannover, working seminar, Hannover. | April 2025 |

“On the rationality problem for hypersurfaces” GLEN Algebraic Geometry Seminar, Sheffield.	April 2025
“On the rationality problem for hypersurfaces” Northern German Algebraic Geometry Seminar, Bielefeld.	January 2025
“Non-Vanishing and hypersurfaces singular along an r -plane” Irrationality, unramified cohomology and hypersurfaces, JAVA, Maison Clément.	July 2024
“On zero-cycles of varieties over Laurent fields” Leibniz University Hannover, AG seminar, Hannover.	April 2024
“Zero-cycles and the étale cycle class map” Leibniz University Hannover, working seminar, Hannover.	November 2023
“Very general (3,3) fivefolds are irrational” (speed talk) Birational Complexity of Algebraic Varieties, Simons Center, Stony Brook.	December 2022

Poster presentations

“On the rationality problem for low degree hypersurfaces” 2025 Summer Research Institute in Algebraic Geometry, Fort Collins.	July 2025
“On the rationality problem for low degree hypersurfaces” Workshop on Algebraic Cycles, Hannover.	October 2024
“On zero-cycles of varieties over Laurent fields” Géométrie Algébrique en Liberté XXXI, Torino.	June 2024
“On zero-cycles of varieties over Laurent fields” Non-Archimedean Geometry, Birational Geometry and Resolution of Sing., Heidelberg.	March 2024

Awards

Doctoral scholarship by Studienstiftung des deutschen Volkes	2023 – 2026
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Teaching Experience

Leibniz University Hannover

Tutorials for Algebra I	Winter 2025
Tutorials for Linear Algebra I	Winter 2024
Tutorials for Topology	Winter 2023
Tutorials for Mathematics for Engineers II	Summer 2023
Tutorials for Linear Algebra I	Winter 2020

Service

Leibniz University Hannover

PhD student representative of the math and physics department	2025 – today
coadvised with S. Schreieder a Bachelor thesis on the Jordan curve theorem	Summer 2024